Chapter 8: Privacy, Ethics and Security

- Privacy
- Ethics
- Security
- Protect Computer Systems

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People in Information System

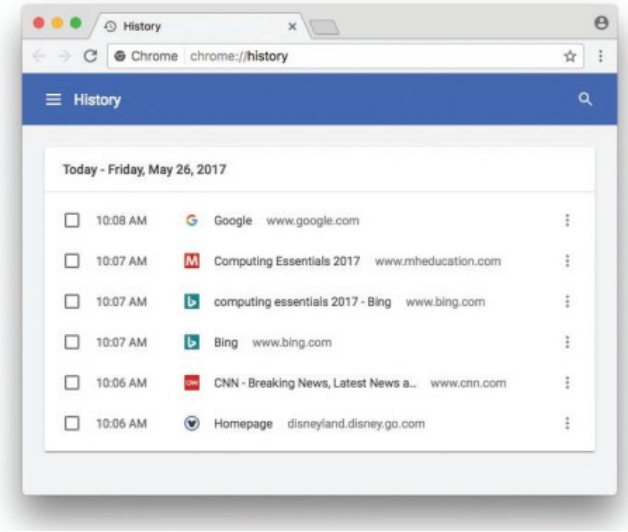
- **Information Systems** consist of people, procedures, software, hardware, data, and the Internet.
- Effective implementation of computer technology involves maximizing its positive effects while minimizing its negative effects. The most significant concerns are:
  - **Privacy**: *What are the threats to personal privacy, and how can we protect ourselves?*
  - **Security**: *How can access to sensitive information be controlled, and how can we secure hardware and software?*
  - **Ethics**: *How do the actions of individual users and companies affect society?*

Privacy

- **Privacy** concerns the collection and use of data about individuals. There are three primary privacy issues:
  - **accuracy** (*who is responsible to ensure data is correct*),
  - **property** (*who owns data*),
  - **access** (*who controls access to data*)
- **Large databases**:
  - Large organizations are constantly compiling information about us. This ever-growing volume of data is often referred to as big data. Information resellers/brokers collect and sell personal data.
  - *Electronic profiles* are compiled from databases to provide highly detailed and personalized descriptions of individuals.
    - *Collecting public, but personally identifying, information*: What if people anywhere in the world could view detailed images of you, or your home?
    - *Spreading information without personal consent*: How would you feel if an employer were using your Facebook, Twitter, to make decisions about hiring, promotion, and firing?
    - *Spreading inaccurate information*: How would you like to be turned down for a home loan because of an error in your credit history?

Privacy

- **Private networks**: Many organizations monitor employee e-mail and computer files using special software called **employee monitoring software**



- **The Internet and the Web**:
  - *Information* stored by browsers includes history files (record sites visited) and temporary Internet files or browser cache.
  - *Cookies* store and track information. Privacy mode ensures that your browsing activity is not recorded.
  - *Spyware* secretly records and reports Internet activities.
    - Keyloggers software records every activity and keystroke.
    - Antispyware detects and removes various privacy threats



Privacy

- **Online Identity**:
  - With the popularity of social networking, blogging, and photo- and video-sharing sites, many people post intimate details of their lives without considering the consequences. This creates an **online identity**.
  - With the archiving and search features of the web, this identity is indefinitely available to anyone who cares to look for it.





Security

- Computer **security** focuses on protecting information, hardware, and software from unauthorized use as well as preventing damage from intrusions, sabotage, and natural disasters
- **Cybercrime**: is a criminal offense involving a computer or network:
  - **Identity theft**—illegal assumption of someone's identity for the purposes of economic gain.
  - **Internet scams**—use Internet to create financial and legal problems for people.
  - **Data manipulation**—unauthorized access of a network to copy files.
  - **Ransomware**—malicious software that encrypts computer data and ransoms owner for access.
  - **Denial of service (DoS) attack**—stops or slows computer network by repeatedly requesting service. Distributed denial of service (DDoS) coordinates several computers

Security

- **Social Engineering**:
  - Social engineering practice of manipulating people to expose private data.
  - Phishing attempts to trick people into believing a fake website is real



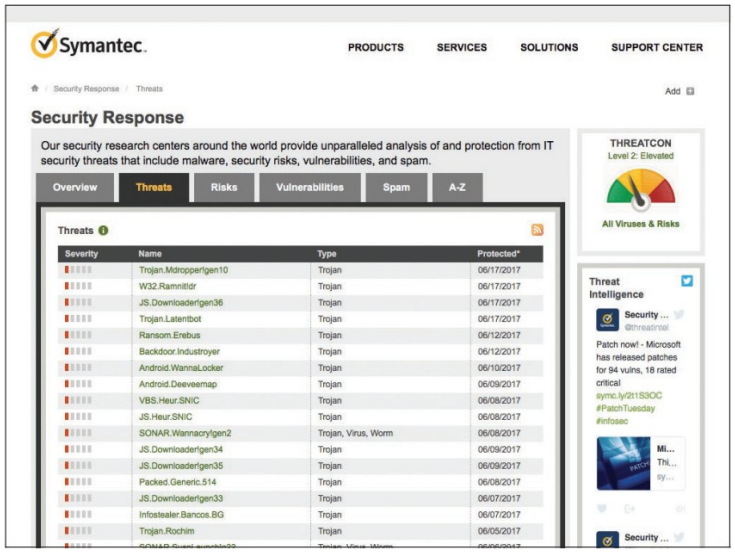
- **Malicious software**: Crackers create and distribute malware (malicious software). Three most common types are
  - *Viruses* migrate through networks and operating systems; can alter/delete files
  - *Worms* repeatedly replicate themselves, clogging computers and networks.
  - *Trojan horses* appear harmless but contain malicious programs



Security

- **Malicious hardware**: Cyber criminals can use computer hardware to steal information:
  - **Zombies**—remotely controlled computers used for malicious purposes; a collection is known as a botnet (robot network).
  - **Rogue Wi-Fi hotspots**—network that appears to be a legitimate free Wi-Fi hot. Once connected, any input by users is captured.
  - **Infected USB flash drives**—USB drives that are often free that contain a virus



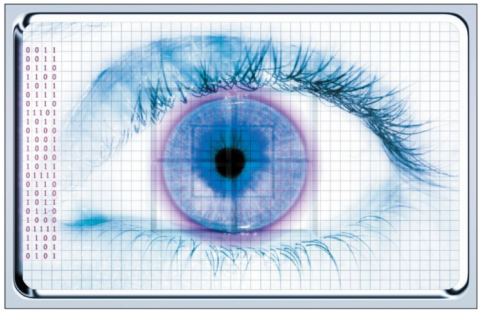


Measures to Protect computer Security

- **Computer Fraud and Abuse Act** makes it a crime for unauthorized persons to view data across state lines using computers; prohibits unauthorized use of computers owned by government or federally insured financial institution.
- Ways to protect computer security:
  - **Access** can be **restricted** through biometric scanning devices and passwords; security suites, firewalls, and password managers help with security tasks
  - **Encrypting** is coding information to make it unreadable except to those who have the *encryption key*
  - **Anticipating disasters** involves physical security, data security, and disaster *recovery* plans
  - **Preventing data loss** involves *protecting* data by screening job applicants, guarding passwords, and auditing and backing up data



Fingerprint scan


Iris scan

Ethics

- ***What do you suppose controls how computers can be used***? You probably think first of **laws**. Of course, that is right, but technology is moving so fast that it is very difficult for our legal system to keep up. The essential element that controls how computers are used today is **ethics.**



- **Ethics** are standards of moral conduct. **Computer ethics** are guidelines for the morally acceptable use of computers in our society.

- **Copyright and digital rights Management**: Copyright is a legal concept that gives content creators the right to control use and distribution of their work

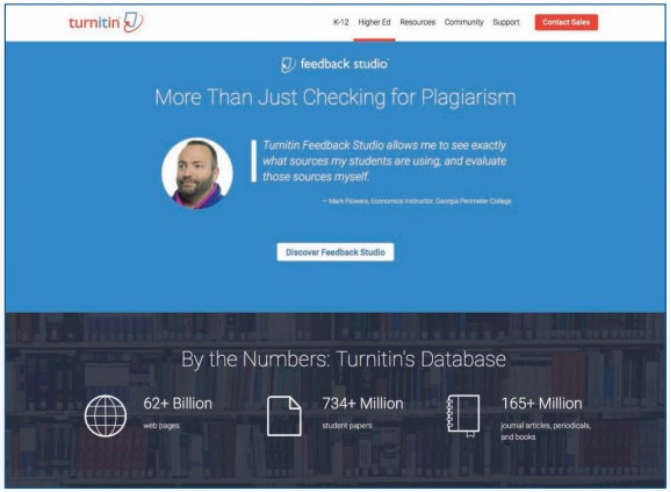
Ethics

- **Software piracy** is the unauthorized copying and distribution of software. Many legal sources for digital media exist, including TV programs that can be watched online, often for free, on TV-network-sponsored sites.



- **Cyberbullying**: the use of the Internet, smartphones, or other devices to send or post content intended to hurt or embarrass another person. (e.g. sending repeated unwanted e-mails)

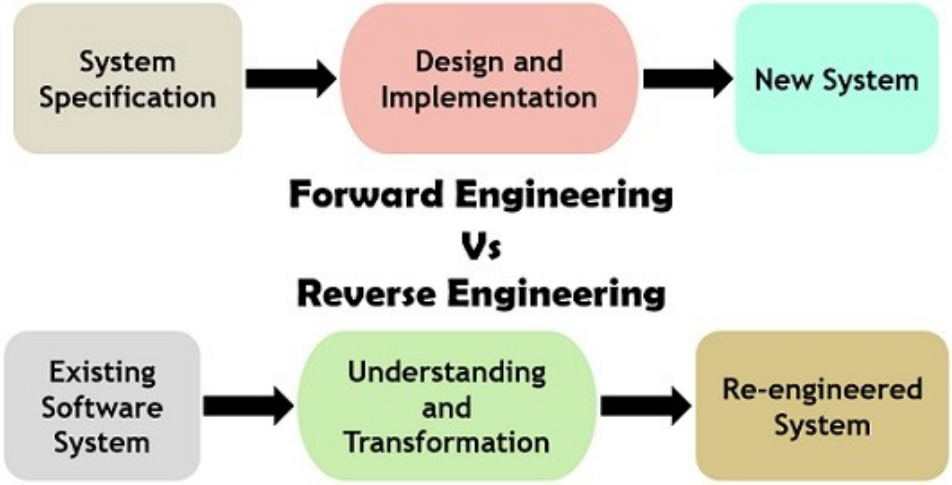
- **Plagiarism:** the illegal and unethical representation of some other person's work and ideas as your own without giving credit to the original sourceIt. (e.g. simply cutting and pasting content from a web page into a report)

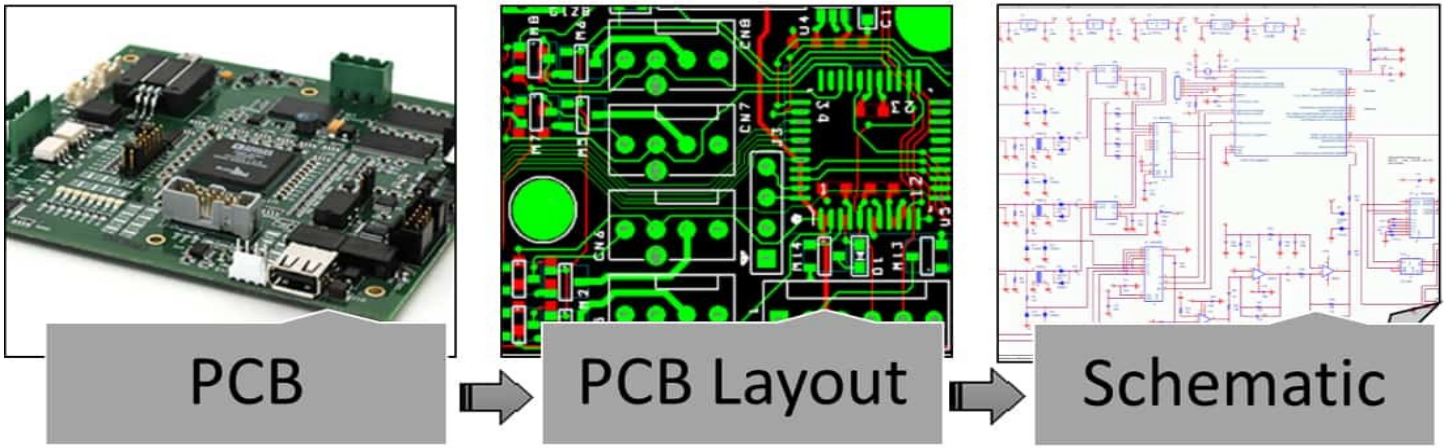


Ethics: Example

❑ Reverse Engineering

- ❖ The process of taking something apart in order to understand it, build a copy of it, or improve it.
- ❖ Applicable for both hardware and software.
- ❖ **Violates** copyright and trade secret laws





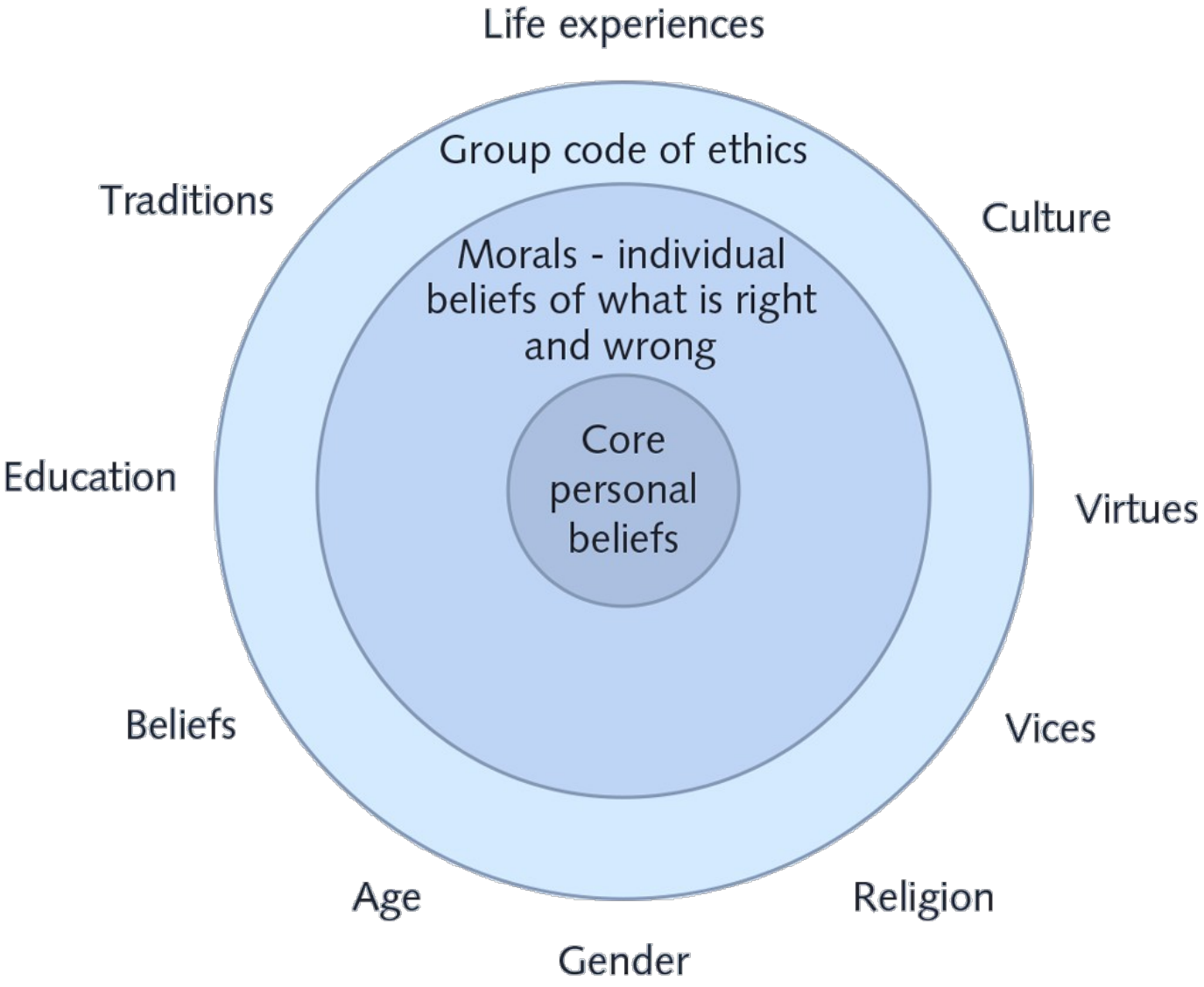
Morals, Ethics, and Laws

Morals	Ethics	Laws
• Personal principles an **individual** uses to determine what is right and what is wrong • Moral acts conform to what an **individual** believes to be the right thing to do	• Code of behavior that is defined by the **group** to which an **individual** belongs	• System of rules, enforced by a set of **institutions**, that tells us what we can and cannot do • Legal acts are acts that **conform** to the law

Morals, Ethics, and Laws



Relationship Between Ethics and Morals



Revision

1. The three primary privacy issues are accuracy, property, and:

a. access
b. ethics
c. ownership
d. security

2. Highly detailed and personalized descriptions of individuals are electronic:

a. cookies
b. histories
c. profiles
d. web bugs

3. Browsers store the locations of sites visited in a:

a. history file
b. menu
c. tool bar
d. firewall

4. The browser mode that ensures your browsing activity is not recorded:

a. detect
b. insert
c. privacy
d. sleep

Revision

Match each numbered item with the most closely related lettered item. Write your answers in the spaces provided

a. accuracy
b. biometric
c. cookies
d. encryption
e. information brokers
f. malware
g. phishing
h. plagiarism
i. spyware
j. zombies

__**1.** Privacy concern that relates to the responsibility to ensure correct data collection.
__**2.** Individuals who collect and sell personal data.
__ **3.** Small data files deposited on your hard disk from websites you have visited.
__ **4.** Wide range of programs that secretly record and report an individual's activities on the Internet.
__ **5.** Malicious programs that damage or disrupt a computer system.
__ **6.** Infected computers that can be remotely controlled.
__ **7.** Used by scammers to trick Internet users with official-looking websites.
__ **8.** A type of scanning device such as fingerprint and iris (eye) scanner.
__ **9.** Process of coding information to make it